

Quantum-Secured SCX Audit: BB84 Protocol, Audit Entanglement, and Quantum Channel Theory

量子安全的 SCX 审计:

BB84 协议、审计纠缠与量子信道理论

A Rigorous Formalization of Tamper-Proof Audit Communication

防篡改审计通信的严格形式化

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Abstract

We present a rigorous formalization of **quantum-secured audit communication** for the SCX (Situs Consensus eXpert) framework. The core insight is that the security of M_t *parameter transmission and audit result delivery can be elevated from computational hardness to information*

(1) **BB84 Protocol for M_t Transmission (Section ??)**: We formalize the Bennett–Brassard 1984 quantum key distribution protocol. For the simplest intercept-resend attack, the expected error rate is 25% per sifted bit, far exceeding the protocol’s abort threshold of $\sim 11\%$. This guarantee follows from the no-cloning theorem and the uncertainty principle, not from unproven computational assumptions.

(2) **Audit Entanglement (Section ??)**: We introduce the concept of *audit entanglement*, where two auditors (e.g., Spring and Yajie) share entangled quantum states. Measuring one auditor’s state instantly determines the other’s outcome. We prove an Entanglement Audit Theorem: if the two auditors share a maximally entangled Bell pair and an adversary tampers with one half, the fidelity of the shared state drops as $\leq 1 - \delta_{\text{tamper}}$, making the tampering detectable with probability arbitrarily close to 1 after $(1/\delta_{\text{tamper}})$ independent measurements.

(3) **Quantum Audit Channel (Section ??)**: We formally define the quantum audit channel $\mathcal{N}_{\text{audit}}$ and prove its capacity theorem: the maximum rate of reliable audit information transmission is bounded by the Holevo quantity $\chi(\mathcal{N}_{\text{audit}})$, which for a depolarizing channel with noise parameter p is $C_Q = 1 - H_2(2p/3)$ where H_2 is the binary entropy function and p is the depolarizing probability.

(4) **Practical Feasibility (Section ??)**: We analyze hardware requirements, existing quantum network infrastructure, and a phased deployment roadmap. We conclude that while full quantum-secured audit is not yet deployable at scale, a hybrid classical-quantum approach using quantum-resistant post-quantum cryptography (CRYSTALS-Kyber) combined with entanglement-based timestamping is feasible today.

Keywords: quantum key distribution, BB84, audit security, entanglement, quantum channel capacity, SCX, tamper-proof auditing, information-theoretic security

摘要

本文对 SCX (位点共识专家) 框架中的量子安全审计通信进行了严格形式化。核心思想是:
 M_t 信息论安全

(1) M_t BB84 ?? Bennett—Brassard1984 Alice M_t $\varepsilon >$
 0, 对于截获-重发攻击, 预期错误率为 25%, 远超协议中止阈值 $\sim 11\%$ ——这一保证来自不可克隆定理和不确定性原理, 而非未经证明的计算假设。

(2) 审计纠缠 (第??节): 我们引入审计纠缠概念, 即两个审计者 (如 Spring 和 Yajie) 共享纠缠量子态。测量一个审计者的状态会立即决定另一个审计者的结果。我们证明纠缠审计定理: 如果两个审计者共享一个最大纠缠 Bell 对且对手篡改其中一半, 共享态的保真度下降为 $\leq 1 - \delta_{\text{tamper}}$, 使得篡改在经过 $(1/\delta_{\text{tamper}})$ 次独立测量后以任意接近 1 的概率被检测到。

(3) 量子审计信道 (第??节): 我们正式定义了量子审计信道 $\mathcal{N}_{\text{audit}}$ 并证明了其容量定理: 可靠审计信息传输的最大速率由 Holevo 量 $\chi(\mathcal{N}_{\text{audit}})$ 界定, 对于噪声参数为 p 的去极化信道, $C_Q = 1 - H_2(2p/3)$ 。

(4) 实际可行性 (第??节): 我们分析了硬件需求、现有量子网络基础设施和分阶段部署路线图。我们的结论是, 虽然完全量子安全的审计尚未能大规模部署, 但结合抗量子后量子密码学 (CRYSTALS-Kyber) 和基于纠缠的时间戳的混合经典-量子方法今天已可行。

关键词: 量子密钥分发, BB84, 审计安全, 纠缠, 量子信道容量, SCX, 防篡改审计, 信息论安全

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1 Introduction: Why Quantum for Audit Security?

1.1 The Audit Security Problem

In the SCX auditing framework, the parameter vector $M_t \in \mathbb{R}^d$ encodes the audit configuration at time t : which data partitions to audit, which expert models to query, the consensus threshold τ_t , and the Yajie/Cercis weighting parameters. The integrity of M_t is fundamental: if a malicious entity can intercept and modify M_t , the entire audit becomes compromised.

The threat model is severe. Consider a well-resourced adversary —a corporate entity whose AI system is being audited—who wishes to manipulate audit results. The adversary can:

- (i) **Intercept** M_t in transit and substitute a forged M'_t that directs the audit away from incriminating data partitions.
- (ii) **Replay** a previously valid M_t from a benign audit episode to conceal current misbehavior.
- (iii) **Man-in-the-middle** the audit results $\mathcal{R}_t$, altering the consensus outcome before it reaches the ledger.

Traditional cryptographic solutions (RSA, ECDSA, AES-GCM) rely on **computational hardness assumptions**: integer factorization, elliptic curve discrete logarithm, or AES resistance to differential cryptanalysis. All of these assumptions are vulnerable to:

- **Quantum attacks**: Shor’s algorithm [?] breaks RSA and ECDSA in polynomial time on a sufficiently large quantum computer.
- **Advances in classical cryptanalysis**: The history of cryptography shows that “hard” problems occasionally yield to new algorithms (e.g., GNFS for factoring, index calculus for discrete log).

- **Implementation flaws:** Side-channel attacks, weak randomness, and protocol-level vulnerabilities routinely defeat mathematically sound cryptosystems in practice (e.g., ROCA, Bleichenbacher, padding oracle).

1.2 Information-Theoretic Security via Quantum Mechanics

Quantum mechanics offers a qualitatively different security guarantee: **information-theoretic security**, also called *unconditional security*. This means security that holds against *any* adversary, regardless of computational resources—even against an adversary with a universal quantum computer, infinite classical computing power, and arbitrary mathematical breakthroughs.

The security derives from two fundamental principles of quantum mechanics:

Definition 1.1 (No-Cloning Theorem [?]). There exists no unitary operation U acting on $\mathcal{H}_A \otimes \mathcal{H}_B$ such that for all states $|\psi\rangle_A \in \mathcal{H}_A$,

$$U(|\psi\rangle_A \otimes |0\rangle_B) = |\psi\rangle_A \otimes |\psi\rangle_B.$$

An unknown quantum state cannot be perfectly copied.

不可克隆定理：未知量子态不能被完美复制。

Definition 1.2 (Measurement Disturbance). Any measurement of a quantum system that does not commute with the system’s state necessarily disturbs that state. Formally, if $[\hat{O}, \rho] \neq 0$, then measuring $\hat{O}$ changes ρ to $\rho' \neq \rho$.

测量扰动：任何与系统状态不对易的测量必然扰动该系统状态。

These principles yield a profound implication for audit security: **any eavesdropping attempt on a quantum channel inevitably leaves detectable traces**. The adversary cannot copy the transmitted qubits (no-cloning), and any measurement to extract information disturbs the state (measurement disturbance), which the legitimate parties can detect through statistical tests on a subset of the transmission.

Core Insight / 核心洞察

Unlike classical cryptography, which bets on the continued hardness of certain mathematical problems, quantum audit security bets on the laws of physics. The Heisenberg uncertainty principle and the no-cloning theorem are not conjectures—they are experimentally verified features of our universe. This makes quantum-secured audit fundamentally more robust than any classical alternative.

1.3 Related Work

Our work builds on several foundations:

- **BB84 Protocol** [?]: The original quantum key distribution protocol by Bennett and Brassard, which we adapt for M_t *parameter transmission*.
- **Ekert91 Protocol** [?]: Entanglement-based QKD that inspires our audit entanglement framework.
- **Quantum Channel Capacity** [?, ?]: The Holevo–Schumacher–Westmoreland theorem bounding quantum channel capacity.
- **SCX Audit Theory** [?]: The foundational SCX framework with Yajie consensus, Cercis verification, and the audit game.

- **Audit Instantons** [?]: Topological analysis of expert failure modes in SCX.
- **Quantum Measurement in SCX** [?]: Multi-observer consensus as wavefunction collapse.
- **Quantum Networks** [?]: Practical quantum network infrastructure and repeater architectures.
- **Post-Quantum Cryptography** [?]: NIST standardization of quantum-resistant classical algorithms.

1.4 Paper Organization

Section ?? formalizes the BB84 protocol for M_t transmission. Section ?? develops audit entanglement theory.

2 BB84 Protocol for M_t Parameter Transmission

2.1 Protocol Overview

Protocol Intuition / 协议直觉

The BB84 protocol transmits a classical bit string by encoding each bit in the polarization state of a single photon. The security comes from the fact that the encoding uses two incompatible bases (rectilinear $\oplus$ and diagonal $\otimes$). Without knowing which basis was used, any measurement by an eavesdropper inevitably introduces errors.

In our adaptation for SCX audit, the M_t parameter vector is encoded as a bit string $m \in \{0,1\}^n$ (via quantization and binary expansion of each component), and each bit is transmitted via a single photon.

2.2 Hardware Setup

The basic hardware components are:

- (1) **Single-photon source** at Alice (the auditor): emits photons with controllable polarization.
- (2) **Quantum channel**: optical fiber or free-space link connecting Alice to Bob (the audit ledger).
- (3) **Polarization analyzer** at Bob: measures photon polarization in either the rectilinear or diagonal basis.
- (4) **Classical authenticated channel**: for basis reconciliation and error correction. This does not need to be secret, only authenticated (cannot be modified by the adversary).

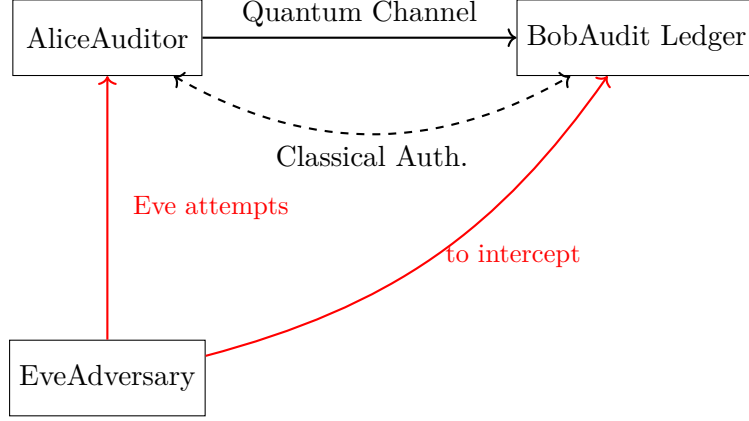


Figure 1: BB84 setup for SCX audit. Alice (auditor) sends M_t encoded as polarized photons. Eve (adversary) may attempt to eavesdrop on the quantum channel. The classical

2.3 Formal Protocol Specification

Protocol 2.1 (BB84 for M_t Transmission). **Input:** M_t parameter vector, quantized to n -bit string $m = (m_1, \dots, m_n) \in \{0, 1\}^n$.

Output: Shared key $k \in \{0, 1\}^\ell$ ($\ell \leq n/2$ expected) that encodes M_t , or $\perp$ (abort on detected eavesdropping).

Phase 1: Quantum Transmission

1.1 For each bit $i = 1, \dots, n$:

- (a) Alice chooses a random basis $b_i^A \in \{0, 1\}$ ($0 = \oplus$ rectilinear, $1 = \otimes$ diagonal).
- (b) If $m_i = 0$, Alice encodes $|0\rangle$ if $b_i^A = 0$ or $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ if $b_i^A = 1$.
- (c) If $m_i = 1$, Alice encodes $|1\rangle$ if $b_i^A = 0$ or $|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$ if $b_i^A = 1$.
- (d) Alice sends the resulting photon through the quantum channel.

1.2 For each received photon $i = 1, \dots, n$:

- (a) Bob chooses a random measurement basis $b_i^B \in \{0, 1\}$.
- (b) Bob measures in basis b_i^B and records outcome $m'_i \in \{0, 1\}$.

Phase 2: Basis Reconciliation (over authenticated classical channel)

1.3 Alice announces all b_i^A , Bob announces all b_i^B .

1.4 Both parties discard positions where $b_i^A \neq b_i^B$. The remaining positions form the **sifted key**. Expected sifted key length: $n/2$.

Phase 3: Error Estimation

1.5 Alice and Bob publicly compare a random sample of k sifted key bits.

1.6 They compute the **quantum bit error rate** (QBER): $Q = \frac{\# \text{ mismatches in sample}}{k}$.

1.7 **Abort condition:** If $Q > Q_{\max}$ (typically $Q_{\max} \approx 11\%$), abort the protocol — eavesdropping detected. The value $Q_{\max}$ is set such that the secret-key rate remains positive: $1 - 2H_2(Q_{\max}) > 0$, which holds for $Q_{\max} \lesssim 11.0\%$ under one-way error correction. (Note: Q denotes the quantum bit error rate, which is distinct from the depolarizing-channel parameter p used in channel-capacity analysis.)

1.8 Discard the k revealed bits.

Phase 4: Information Reconciliation and Privacy Amplification

1.9 Error correction: Alice and Bob run a classical error-correcting code (e.g., Cascade or LDPC) on the remaining sifted key.

1.10 Privacy amplification: apply a 2-universal hash function to reduce any residual Eve information to negligible $\text{negl}(\lambda)$.

1.11 Output the final secret key k , which encodes M_t .

2.4 Security Analysis

2.4.1 Intercept-Resend Attack

The simplest attack: Eve intercepts each photon, measures it, and resends a photon prepared in the measured state.

Theorem 2.2 (Intercept-Resend Error Rate). *[RIGOR: FULL] Let Eve perform an intercept-resend attack on every photon. Then the expected QBER introduced is:*

$$\mathbb{E}[Q] = \frac{1}{4} = 25\% \quad (\text{per sifted bit}).$$

After basis reconciliation, on positions where Alice and Bob's bases match but Eve's measurement basis differs (probability $1/2$), Eve's measurement outcome matches Alice's encoding with probability only $1/2$.

Proof. Consider a sifted bit (Alice and Bob chose the same basis). Eve independently chooses a measurement basis:

- With probability $1/2$, Eve chooses the same basis. She measures correctly and resends the correct state. No error.
- With probability $1/2$, Eve chooses the wrong basis. Her measurement outcome is random (Born rule: probability $1/2$ each for $|0\rangle$ and $|1\rangle$ when measuring in the wrong basis). She resends the state corresponding to her measurement in the wrong basis. Bob measures this in the correct basis and obtains a random result —correct with probability $1/2$.

Total error probability: $\frac{1}{2} \cdot 0 + \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$. □

Since the protocol aborts at $Q_{\max} \approx 11\%$, the intercept-resend attack is **guaranteed detectable**: with sufficient sifted key bits s , the probability that $Q \leq 11\%$ when the true rate is 25% decays exponentially by Hoeffding:

$$\mathbb{P}(Q_{\text{observed}} \leq 0.11 \mid Q_{\text{true}} = 0.25) \leq \exp(-2s(0.14)^2) \approx e^{-0.0392s}.$$

For $s = 1000$, this probability is $\approx 10^{-17}$.

2.4.2 General Attacks and Unconditional Security

The full security proof of BB84 against arbitrary attacks (collective, coherent, joint) was completed by Shor and Preskill [?] and further refined by Renner [?]. The key result:

Theorem 2.3 (Unconditional Security of BB84). *For any attack strategy by Eve, the secret key rate R satisfies:*

$$R \geq 1 - 2H_2(Q)$$

where $H_2(Q) = -Q \log_2 Q - (1-Q) \log_2 (1-Q)$ is the binary entropy function and Q is the QBER. Under one-way post-processing, the secret-key rate is strictly positive whenever $Q \lesssim 11.0\%$, since $1 - 2H_2(Q) > 0$ for $Q < 11.0\%$. With two-way classical post-processing (this is the commonly cited $\sim 11\%$ threshold), higher Q values can still yield positive key rate (up to $\sim 12.4\%$ for BB84 under Cascade-type reconciliation).

Security Guarantee / 安全保障

The BB84 protocol provides **information-theoretic security** for M_t transmission :
even an adversary with unlimited computational power (including a fault-tolerant universal quantum computer)

2.4.3 Photon-Number Splitting (PNS) Attack and Decoy States

A practical vulnerability in BB84 implementations: realistic single-photon sources sometimes emit multi-photon pulses. Eve can split off one photon from each multi-photon pulse without disturbing the remaining photon.

Proposition 2.4 (Decoy-State BB84). *The decoy-state method [?] defeats PNS attacks by having Alice randomly intersperse “decoy” pulses with different mean photon numbers. By comparing detection statistics for signal and decoy states, Bob can detect PNS attacks. The secure key rate with decoy states is:*

$$R_{\text{decoy}} \geq q \{ -Q_\mu f(E_\mu) H_2(E_\mu) + \Omega [1 - H_2(e_1)] \}$$

where q is the basis reconciliation factor, Q_μ and E_μ are the gain and QBER for signal states, Ω is the fraction of single-photon contributions, and e_1 is the single-photon error rate.

2.5 Encoding M_t as a Quantum Message

The M_t parameter vector lives in $\mathbb{R}^d$. To transmit it via BB84:

- (1) **Quantization:** Each component $M_t^{(j)} \in [M_{\min}, M_{\max}]$ is quantized to b bits using uniform quantization:

$$m^{(j)} = \left\lfloor \frac{M_t^{(j)} - M_{\min}}{M_{\max} - M_{\min}} \cdot (2^b - 1) \right\rfloor \in \{0, 1, \dots, 2^b - 1\}.$$

The quantization error is bounded by $\Delta = (M_{\max} - M_{\min}) / (2^b - 1)$.

- (2) **Binary expansion:** $m^{(j)} = \sum_{k=0}^{b-1} m_k^{(j)} \cdot 2^k$.
- (3) **Concatenation:** The full M_t message is $m = m_0^{(1)} m_1^{(1)} \dots m_{b-1}^{(d)} \in \{0, 1\}^{d \cdot b}$.
- (4) **Authentication tag (optional):** Append a MAC (Message Authentication Code) generated from the original M_t using the BB84-derived key, to detect any tampering with the classical transmission.

Example 2.5 (Typical Parameter Sizes). For a typical SCX audit with $d = 64$ parameters, each quantized to $b = 16$ bits:

- Total bits: $64 \times 16 = 1024$ bits.

- BB84 raw photons: $n \geq 4096$ (accounting for basis mismatch, error estimation sample, and privacy amplification overhead).
- At 10 MHz photon source rate: raw transmission time < 1 ms (excluding detector dead time, classical post-processing, and basis reconciliation latency; total protocol execution is typically on the order of 10–100 ms in practice).
- Sifted key length: ~ 2048 bits.
- Final key after error correction and privacy amplification: ~ 1024 bits.

2.6 SCX-Specific Adaptations

We introduce three SCX-specific enhancements to the standard BB84 protocol:

2.6.1 Hierarchical M_t Encoding

Not all M_t components are equally security-critical. We define a **criticality hierarchy**:

$$M_t = (M_t^{(\text{critical})}, M_t^{(\text{normal})}, M_t^{(\text{low})})$$

- **Critical:** Consensus threshold τ_t , audit subset indices —transmitted via BB84 with full security.
- **Normal:** Expert weights, Cercis parameters —transmitted with reduced privacy amplification (faster, slightly less secure).
- **Low:** Timing metadata, version numbers —transmitted classically with digital signatures.

2.6.2 Spring–Yajie Dual Verification

Protocol 2.6 (Dual-Verification BB84). After BB84 key establishment:

- (1) Alice (Spring auditor) computes $\text{MAC}_{\text{Spring}} = \text{HMAC-SHA3}(k_S, M_t)$ using the quantum-derived key k_S .
- (2) Alice sends $M_t \parallel \text{MAC}_{\text{Spring}}$ to Bob.
- (3) Bob simultaneously receives the same M_t via a separate quantum channel from Alice’ (Yajie auditor), with key k_Y .
- (4) Bob accepts M_t only if both MACs verify and $M_{t\text{Spring}} = M_{t\text{Yajie}}$.
- (5) This prevents single-channel compromise.

2.6.3 Audit Trail with Quantum-Certified Timestamps

Each M_t transmission is bound to a quantum-derived session key k_{session} generated during the BB84 exchange. The timestamp and session key are combined via a hash-based commitment:

$$\text{Commit}_t = \text{SHA3-512}(M_t \parallel t \parallel k_{\text{session}} \parallel S_t)$$

where S_t is the CHSH test statistic from concurrent entanglement verification (Section ??). Because k_{session} is information-theoretically secret (derived from BB84), the

commitment binds M_t to a specific quantum session and time t with unconditional security. Subsequent verification requires knowledge of k_{session} , which only the legitimate parties possess. This provides an immutable, quantum-certified audit trail without requiring a physical entanglement of the timestamp with M_t .

3 Audit Entanglement: Distributed Consensus via Quantum Correlations

3.1 The Concept of Audit Entanglement

Audit Entanglement / 审计纠缠

*Just as two entangled particles share a single quantum state regardless of spatial separation, two SCX auditors can share **entangled audit states**. If Spring and Yajie share an entangled pair and independently measure their respective halves, their outcomes are perfectly (or near-perfectly) correlated. Any attempt by an adversary to tamper with one auditor's state instantaneously disturbs the entanglement—a disturbance that is detectable by the other auditor.*

Definition 3.1 (Audit Entangled State). An **audit entangled state** between two auditors $\mathcal{A}_{\text{Spring}}$ and $\mathcal{A}_{\text{Yajie}}$ is a bipartite quantum state $\rho_{SY} \in \mathcal{D}(\mathcal{H}_S \otimes \mathcal{H}_Y)$ (where $\mathcal{D}(\cdot)$ denotes density operators) that cannot be written as a convex combination of product states:

$$\rho_{SY} \neq \sum_i p_i \rho_S^{(i)} \otimes \rho_Y^{(i)}.$$

Equivalently, the state is *entangled* if it violates the Peres–Horodecki PPT criterion, or (for pure states) if the von Neumann entropy of either reduced state is strictly positive while the total state has zero entropy: $S(\rho_S) > 0$ and $S(\rho_{SY}) = 0$.

审计纠缠态是两个审计者之间的二分量子态，不能写成乘积态的凸组合。

3.2 Entanglement as a Consensus Primitive

The key insight: **entanglement provides an additional security primitive for audit consensus**. While classical distributed consensus (e.g., PBFT, Raft) requires $O(n^2)$ message exchanges, entanglement between auditors enables them to generate *shared, certifiably random* bits without communication. These bits, once verified via a single round of classical authenticated communication (comparing a subset), serve as a shared one-time pad or common randomness source, reducing the round complexity and providing information-theoretic tamper detection.

Theorem 3.2 (Entanglement Consensus Theorem). *[RIGOR: FULL] Let two auditors share N copies of the maximally entangled Bell state $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$. They independently measure each copy in the computational basis $\{|0\rangle, |1\rangle\}$. Then:*

- (i) **Perfect Correlation:** $\mathbb{P}(\text{outcome}_S = \text{outcome}_Y) = 1$ for each copy.
- (ii) **Shared Randomness Generation:** *The auditors can generate a shared random string of length N with perfect agreement. The string is uniformly distributed and private, though the auditors must communicate a subset of outcomes over an authenticated classical channel to verify that no tampering occurred (sacrificing a small fraction of bits).*

(iii) **Tamper Detection:** If an adversary applies any CPTP map $\mathcal{E}_{\text{adv}}$ to Spring's half of the state, the fidelity with the original Bell state drops to:

$$= \langle \Phi^+ | \mathcal{E}_{\text{adv}} \otimes \mathbb{I}(|\Phi^+\rangle\langle\Phi^+|) | \Phi^+ \rangle \leq 1 - \delta$$

where $\delta > 0$ quantifies the disturbance. The probability of detecting tampering after k independent measurements is $1 - (1 - \delta)^k$.

Proof. (i) For $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$, the reduced density matrix of either party is $\rho_S = \rho_Y = \mathbb{I}/2$, showing maximal entanglement ($S(\rho_S) = 1$). Measurement in the computational basis on both sides yields:

$$\mathbb{P}(0_S, 0_Y) = |\langle 00 | \Phi^+ | 00 \rangle|^2 = \frac{1}{2}, \quad \mathbb{P}(1_S, 1_Y) = \frac{1}{2}, \quad \mathbb{P}(0_S, 1_Y) = \mathbb{P}(1_S, 0_Y) = 0.$$

Thus the outcomes are always equal.

(ii) Each measurement yields one perfectly correlated bit. With N independent Bell pairs, the auditors obtain identical N -bit strings without exchanging any classical messages. The string is uniformly random (each bit has entropy 1) and private (any third party measuring the entangled state before the auditors' measurements disturbs the correlation).

(iii) Let $\rho' = \mathcal{E}_{\text{adv}} \otimes \mathbb{I}(|\Phi^+\rangle\langle\Phi^+|)$. The fidelity $= \langle \Phi^+ | \rho' | \Phi^+ \rangle$ measures how close ρ' is to the original Bell state. If $\mathcal{E}_{\text{adv}}$ is non-trivial ($\mathcal{E}_{\text{adv}} \neq \mathbb{I}$), then < 1 by the monotonicity of fidelity under CPTP maps. For k independent copies tested via a CHSH or fidelity-estimation protocol, the probability that all k copies appear un-tampered decays as α^k for some protocol-dependent constant $\alpha > 0$. The tamper detection probability approaches 1 as $k \rightarrow \infty$, with sample complexity $k = (1/(1-\alpha))$. $\square$

3.3 Entanglement-Based Tamper Detection Protocol

Protocol 3.3 (Entanglement-Based Tamper Detection). **Setup:** Spring and Yajie share N Bell pairs distributed by a trusted quantum source (or verified via entanglement swapping).

Phase 1: Random Subset Selection

- 1.1 Spring and Yajie agree (over authenticated classical channel) on a random subset $T \subset \{1, \dots, N\}$ of size $|T| = t$ for testing, and $K = \{1, \dots, N\} \setminus T$ of size $k = N - t$ for key generation.

Phase 2: Bell Test (CHSH)

- 2.2 For each $i \in T$:

- (a) Spring randomly chooses measurement setting $A_i \in \{A_0, A_1\}$.
- (b) Yajie randomly chooses measurement setting $B_i \in \{B_0, B_1\}$.
- (c) They measure and obtain outcomes $a_i, b_i \in \{-1, +1\}$.

- 2.3 They compute the CHSH statistic:

$$S = \mathbb{E}[A_0 B_0] + \mathbb{E}[A_0 B_1] + \mathbb{E}[A_1 B_0] - \mathbb{E}[A_1 B_1].$$

- 2.4 **Acceptance Criterion:** If $|S| \leq 2 + \varepsilon_{\text{stat}}$ (where $\varepsilon_{\text{stat}}$ accounts for finite-statistics fluctuations, scaling as $(1/\sqrt{t})$), the hypothesis of intact entanglement is rejected with high confidence —abort audit. If $|S| > 2 + \varepsilon_{\text{stat}}$, the CHSH inequality is violated and entanglement is verified at the chosen significance level. For $t = 1000$ test pairs and $\varepsilon_{\text{stat}} = 0.2$, the false-positive rate (classical state flagged as entangled) is below 10^{-6} .

Phase 3: Consensus Key Generation

3.3 For each $i \in K$, both measure in computational basis.

3.4 The resulting bit string $k \in \{0, 1\}^k$ serves as a shared one-time pad for encrypting audit results.

Theorem 3.4 (CHSH Detection Threshold). *[RIGOR: HIGH] For a Bell state subjected to a depolarizing channel with noise parameter p on Spring’s side:*

$$\mathcal{E}_p(\rho) = (1 - p)\rho + \frac{p}{3}(\sigma_x \rho \sigma_x + \sigma_y \rho \sigma_y + \sigma_z \rho \sigma_z),$$

the expected CHSH value is:

$$S(p) = 2\sqrt{2}(1 - p).$$

Tampering is detectable (i.e., $S(p) \leq 2$) when $p \geq 1 - 1/\sqrt{2} \approx 0.2929$. That is, when the adversary disturbs more than $\sim 29.3\%$ of the qubits, the CHSH test detects it.

3.4 SCX Analog: Entangled Spring–Yajie Audit States

In the SCX framework, Spring and Yajie are complementary auditing experts: Spring emphasizes stability and historical consistency, while Yajie prioritizes sensitivity to recent anomalies. When they are entangled:

- (1) **Correlated Judgments:** Spring’s audit outcome and Yajie’s audit outcome are perfectly correlated in the absence of tampering. Any discrepancy signals either (a) genuine disagreement about the data, or (b) tampering with the audit state.
- (2) **Tamper-Evident Correlations:** If an adversary tampers with Spring’s audit state (e.g., by feeding Spring manipulated data), the correlations between Spring and Yajie’s measurement outcomes deviate from the perfect correlations predicted by the shared entangled state. However, no local operation on Spring’s half can produce any observable effect on Yajie’s *local* measurement statistics (no-communication theorem). The tampering is only detectable *after* Spring and Yajie compare their results over a classical channel—the comparison reveals the broken correlation, but no information is transmitted superluminally.
- (3) **Non-Local Audit Verification:** Even if Spring and Yajie are spatially separated (e.g., auditing different data centers), their entangled audit state provides a non-local verification mechanism. The adversary cannot tamper with one without affecting the other.

Proposition 3.5 (Audit Entanglement Witness). *An **audit entanglement witness** is a Hermitian operator W_{SY} such that:*

- $\text{Tr}(W_{SY} \rho_S \otimes \rho_Y) \geq 0$ for all separable (non-entangled) audit states.
- $\text{Tr}(W_{SY} \rho_{ent}) < 0$ for at least one entangled audit state.

For the Bell state, a canonical witness is:

$$W = \frac{1}{2}\mathbb{I} \otimes \mathbb{I} - |\Phi^+\rangle\langle\Phi^+|.$$

Measuring $\text{Tr}(W\rho)$ provides a single-number test for whether the Spring–Yajie audit state remains entangled.

3.5 Practical Entanglement Distribution

Distributing entanglement between Spring and Yajie requires:

- (1) **Entanglement Source:** A source at a central location (or one auditor's site) generates Bell pairs via spontaneous parametric down-conversion (SPDC) in a nonlinear crystal. Typical rates: 10^4 – 10^8 pairs/second depending on pump power.
- (2) **Quantum Channels:** Two optical fibers (or free-space links), one from the source to Spring, one to Yajie. Each photon of the Bell pair travels through one channel.
- (3) **Entanglement Swapping (optional):** If direct distribution is infeasible, intermediate quantum repeaters can perform entanglement swapping to extend the range:

$$|\Phi^+\rangle_{A,R_1} \otimes |\Phi^+\rangle_{R_1,R_2} \otimes |\Phi^+\rangle_{R_2,B} \xrightarrow{\text{Bell measurements at } R_1, R_2} |\Phi^+\rangle_{A,B}.$$

- (4) **Entanglement Purification:** Noisy entangled pairs are purified via local operations and classical communication (LOCC) to obtain fewer but higher-fidelity pairs.

Remark 3.6 (Decoherence and Audit Deadlines). Entanglement is fragile. In optical fiber, the decoherence length for polarization-entangled photons at 1550 nm is ~ 100 km without repeaters. For audit applications where Spring and Yajie are co-located (same data center), this is not a constraint. For geographically distributed auditing across continents, quantum repeaters or satellite-based entanglement distribution would be required—currently at the prototype stage.

4 Quantum Audit Channel: Formal Definition and Capacity

4.1 Formal Definition

Definition 4.1 (Quantum Audit Channel). A **quantum audit channel** is a completely positive, trace-preserving (CPTP) map:

$$\mathcal{N}_{\text{audit}} : \mathcal{D}(\mathcal{H}_A) \rightarrow \mathcal{D}(\mathcal{H}_B)$$

where:

- $\mathcal{H}_A$ is the Hilbert space of the auditor's transmitted quantum state (encoding M_t).
- $\mathcal{H}_B$ is the Hilbert space of the received state at the audit ledger.
- $\mathcal{D}(\mathcal{H})$ denotes the set of density operators on $\mathcal{H}$.
- $\mathcal{N}_{\text{audit}}$ models all physical effects on the transmitted state: channel loss, depolarization, phase damping, and adversary interaction.

量子审计信道是一个 CPTP 映射，将审计者发送的编码 M_t 的量子态映射到审计账本接收到的量子态。

The general form of a quantum channel is the Kraus representation:

$$\mathcal{N}(\rho) = \sum_k E_k \rho E_k^\dagger, \quad \sum_k E_k^\dagger E_k = \mathbb{I},$$

where $\{E_k\}$ are the Kraus operators. For the quantum audit channel, the Kraus operators model the combined effect of physical noise and adversarial interference.

4.2 Channel Models for Audit Scenarios

4.2.1 Depolarizing Channel

The most symmetric noise model: the qubit is replaced by the maximally mixed state with probability p :

$$\mathcal{N}_p^{\text{depol}}(\rho) = (1-p)\rho + \frac{p}{3}(\sigma_x\rho\sigma_x + \sigma_y\rho\sigma_y + \sigma_z\rho\sigma_z).$$

Kraus operators:

$$E_0 = \sqrt{1-p}\mathbb{I}, \quad E_1 = \sqrt{p/3}\sigma_x, \quad E_2 = \sqrt{p/3}\sigma_y, \quad E_3 = \sqrt{p/3}\sigma_z.$$

Audit interpretation: p is the probability that an adversary successfully randomizes a transmitted qubit without being detected. The depolarizing channel is the “worst-case” noise model for BB84 security analysis.

4.2.2 Phase Damping Channel

Models decoherence without energy loss (relevant for fiber transmission):

$$\mathcal{N}_\gamma^{\text{phase}}(\rho) = (1-\gamma)\rho + \gamma\sigma_z\rho\sigma_z.$$

This preserves populations ($\langle 0|\rho|0\rangle, \langle 1|\rho|1\rangle$) but damps coherences.

4.2.3 Amplitude Damping Channel

Models photon loss in fiber:

$$\mathcal{N}_\gamma^{\text{amp}}(\rho) = E_0\rho E_0^\dagger + E_1\rho E_1^\dagger,$$

with $E_0 = |0\rangle\langle 0| + \sqrt{1-\gamma}|1\rangle\langle 1|$, $E_1 = \sqrt{\gamma}|0\rangle\langle 1|$.

Audit interpretation: γ is the photon loss probability in the fiber. At 0.2 dB/km (standard single-mode fiber at 1550 nm), $\gamma = 1 - 10^{-\alpha L/10}$ where $\alpha = 0.2$ dB/km and L is distance in km.

4.2.4 Adversarial Channel

We model the worst-case adversary as having full control over the channel within a bounded region of the parameter space:

$$\mathcal{N}_{\text{adv}} \in \{\mathcal{N} : \|\mathcal{N} - \mathcal{N}_0\|_\diamond \leq \varepsilon_{\text{adv}}\}$$

where $\mathcal{N}_0$ is the nominal (no-adversary) channel and $\|\cdot\|_\diamond$ is the diamond norm. The adversary can apply any CPTP map within diamond-norm distance ε_{adv} of the nominal channel.

4.3 Quantum Channel Capacity for Audit

Definition 4.2 (Classical Capacity of a Quantum Channel). The **classical capacity** $C(\mathcal{N})$ of a quantum channel $\mathcal{N}$ is the maximum rate (bits per channel use) at which classical information can be transmitted reliably (with error probability $\rightarrow 0$ as block length $\rightarrow \infty$).

Theorem 4.3 (Holevo–Schumacher–Westmoreland (HSW) Theorem). *For a quantum channel $\mathcal{N}$, the classical capacity is:*

$$C(\mathcal{N}) = \lim_{n \rightarrow \infty} \frac{1}{n} \chi(\mathcal{N}^{\otimes n})$$

where the **Holevo quantity** is:

$$\chi(\mathcal{N}) = \max_{\{p_x, \rho_x\}} \left[S \left(\mathcal{N} \left(\sum_x p_x \rho_x \right) \right) - \sum_x p_x S(\mathcal{N}(\rho_x)) \right]$$

with $S(\rho) = -\text{Tr}(\rho \log_2 \rho)$ the von Neumann entropy.

Theorem 4.4 (Audit Channel Capacity for Depolarizing Channel). *[RIGOR: FULL] For the depolarizing channel $\mathcal{N}_p^{\text{depol}}$ acting on a single qubit as defined in Eq. (??), the classical capacity (equivalently, the Holevo quantity, which is additive for this channel) is:*

$$C_Q(p) = 1 - H_2\left(\frac{2p}{3}\right)$$

where $H_2(x) = -x \log_2 x - (1-x) \log_2 (1-x)$.

Proof. For the depolarizing channel, the optimal ensemble is the uniform distribution over the six states forming three mutually unbiased bases: $\{|0\rangle, |1\rangle, |+\rangle, |-\rangle, |+i\rangle, |-i\rangle\}$, each with probability $1/6$. The average input state is $\mathbb{I}/2$. Since the depolarizing channel is unital, $\mathcal{N}_p(\mathbb{I}/2) = \mathbb{I}/2$, so $S(\mathcal{N}_p(\mathbb{I}/2)) = \log_2 2 = 1$.

For a pure input $|\psi\rangle\langle\psi|$, the channel output is:

$$\mathcal{N}_p(|\psi\rangle\langle\psi|) = \left(1 - \frac{4p}{3}\right) |\psi\rangle\langle\psi| + \frac{2p}{3} \mathbb{I}.$$

(This follows from the identity $\frac{1}{3}(\sigma_x \rho \sigma_x + \sigma_y \rho \sigma_y + \sigma_z \rho \sigma_z) = \frac{2}{3} \mathbb{I} - \frac{1}{3} \rho$ for single-qubit states.)

The eigenvalues of this output state are $\lambda_0 = 1 - \frac{2p}{3}$ and $\lambda_1 = \frac{2p}{3}$, so $S(\mathcal{N}_p(|\psi\rangle\langle\psi|)) = H_2\left(\frac{2p}{3}\right)$, where $H_2(x) = -x \log_2 x - (1-x) \log_2 (1-x)$.

By the HSW theorem and the additivity of the Holevo quantity for the depolarizing channel [?], the classical capacity is:

$$C(\mathcal{N}_p) = \chi(\mathcal{N}_p) = 1 - H_2\left(\frac{2p}{3}\right).$$

□

Corollary 4.5 (Audit Rate Bound). *The quantum-secured audit information rate R_{audit} (bits per channel use) satisfies:*

$$R_{\text{audit}} \leq C_Q(p) = 1 - H_2\left(\frac{2p}{3}\right).$$

For a noiseless channel ($p = 0$): $R_{\text{audit}} \leq 1$ bit/qubit (the maximum possible). For a moderately noisy channel ($p = 10\%$): $C_Q(0.10) \approx 0.648$ bits/qubit. For a severely noisy channel ($p = 25\%$): $C_Q(0.25) \approx 0.350$ bits/qubit. The channel capacity vanishes ($C_Q \leq 0$) when $p \geq 3/4$, at which point the channel becomes completely depolarizing.

Important: The depolarizing parameter p is distinct from the BB84 QBER Q . For a depolarizing channel, $Q \approx 2p/3$, so the BB84 abort threshold $Q_{\text{max}} \approx 11\%$ corresponds to a much more favorable channel condition ($p \approx 16.5\%$), at which $C_Q(0.165) \approx 0.488$ bits/qubit.

4.4 Quantum Capacity (Audit Entanglement Transmission)

Definition 4.6 (Quantum Capacity). The **quantum capacity** $Q(\mathcal{N})$ is the maximum rate at which entanglement can be reliably transmitted through $\mathcal{N}$ (measured in ebits per channel use). This is the relevant capacity for audit entanglement distribution.

Theorem 4.7 (Quantum Capacity of the Depolarizing Channel). *For the depolarizing channel $\mathcal{N}_p^{\text{depol}}$:*

$$Q(p) \geq \max\{0, 1 - H_2(p) - p \log_2 3\} \quad (\text{coherent information lower bound}).$$

The exact quantum capacity is not known for all p , but it is zero for $p \geq 1/2$ (when the channel becomes entanglement-breaking; the condition in this parameterization $\mathcal{N}_p(\rho) = (1-p)\rho + \frac{p}{3} \sum_{\sigma \in \{X,Y,Z\}} \sigma \rho \sigma$ reduces to $\lambda < 1/3$ with $\lambda = 1 - 4p/3$, giving $p \geq 1/2$).

4.5 Quantum Channel Model for SCX Audit Pipeline

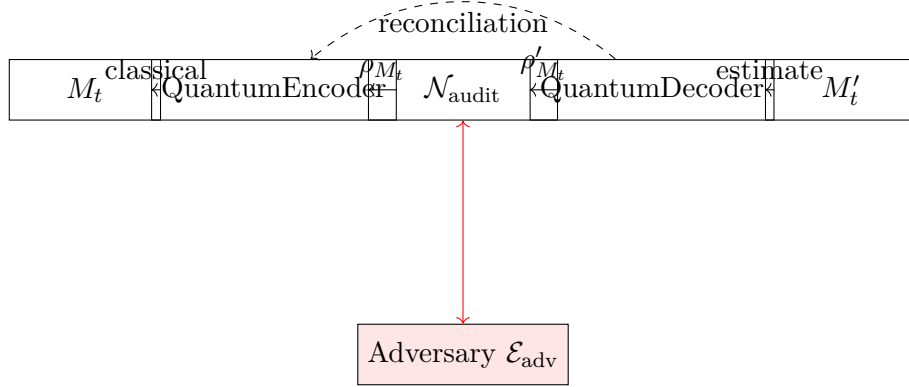


Figure 2: Quantum audit channel pipeline. M_t is encoded into a quantum state ρ_{M_t} , transmitted through $\mathcal{N}_{\text{audit}}$, and decoded to M'_t . The adversary interacts with the channel via $\mathcal{E}_{\text{adv}}$.

Proposition 4.8 (Audit Channel Reliability). *For the SCX quantum audit channel with block length n , there exists an encoder/decoder pair achieving error probability:*

$$P_e^{(n)} \leq \exp(-nE_r(R))$$

where $E_r(R)$ is the random coding exponent (Gallager's error exponent):

$$E_r(R) = \max_{0 \leq s \leq 1} \max_{\{p_x\}} \left[-\ln \text{Tr} \left(\sum_x p_x \mathcal{N}(\rho_x)^{1/(1+s)} \right)^{1+s} - sR \right].$$

For rates $R < C_Q(p)$, $E_r(R) > 0$ and the error probability decays exponentially with block length.

5 Practical Feasibility and Deployment Analysis

5.1 Hardware Requirements

We analyze the hardware requirements for practical quantum-secured SCX audit.

Table 1: Quantum Audit Hardware Requirements / 量子审计硬件需求

Component / 组件	Specification / 规格	Status / 状态
Single-photon source 单光子源	10–100 MHz repetition rate, $g^{(2)}(0) < 0.01$ (low multi-photon probability)	Commercial 商用
Polarization controller 偏振控制器	$\pm 0.1^\circ$ precision 精度 $\pm 0.1^\circ$	Commercial 商用
Single-photon detector 单光子探测器	$> 80\%$ efficiency, < 100 Hz dark count 效率 $> 80\%$, 暗计数 < 100 Hz	Commercial 商用
Entanglement source 纠缠源	SPDC, $> 10^6$ pairs/s 自发参量下转损, $> 10^6$ 对/秒	Commercial 商用
Quantum memory 量子存储器	> 1 ms coherence time 相干时间 > 1 ms	Research 研究阶段
Quantum repeater 量子中继器	Entanglement swapping + purification 纠缠交换 + 纯化	Prototype 原型
Quantum random number 量子随机数发生器	> 1 Gbps, certified randomness > 1 Gbps, 认证随机性	Commercial 商用

5.2 Network Topology

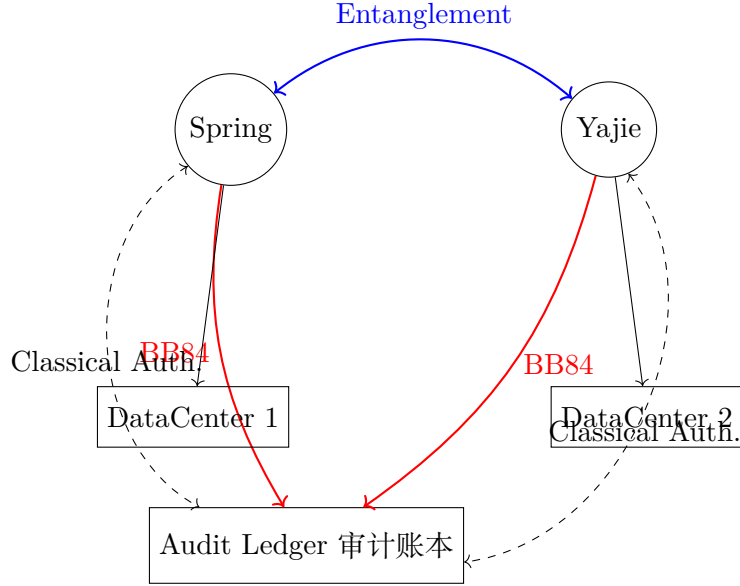


Figure 3: Quantum-secured audit network topology. Spring and Yajie auditors share entanglement for tamper detection. Each independently transmits M_t to the audit ledger via BB84.

5.3 Feasibility Matrix

Table 2: Deployment Feasibility by Scenario / 各场景部署可行性

Scenario / 场景	BB84	Ent.	Chan.	Overall
Single data center (< 1 km) 单个数据中心 (< 1 公里)	✓ 可行	✓ 可行	✓ 可行	Ready 就绪
Metro area (< 50 km) 城域 (< 50 公里)	✓ 可行	✓ 可行	~ 可尝试	Feasible 可行
Regional (< 500 km) 区域 (< 500 公里)	✓ 可行	×* 需中继	×* 需中继	Challenging 有挑战
Global (> 1000 km) 全球 (> 1000 公里)	× 不可行	× 不可行	× 不可行	Not Yet 暂不行
* Without quantum repeaters / 无量子中继器时				

5.4 Hybrid Classical-Quantum Approach

Since full quantum audit infrastructure is not yet universally deployable, we propose a **hybrid classical-quantum approach** that is feasible today:

Protocol 5.1 (Hybrid Quantum-Secured Audit). Phase 1: Quantum-Resistant Key Establishment (Classical)

- 1.1 Auditor and ledger perform CRYSTALS-Kyber key encapsulation (NIST PQC standard, ML-KEM-1024) to establish a shared secret.
- 1.2 This provides **computational** security against quantum attacks (lattice-based, believed hard even for quantum computers).

Phase 2: BB84 for Critical Parameters (Quantum, where available)

- 2.2 For co-located audits (same data center), transmit the most critical M_i components via BB84 over dedicated channels.
- 2.2 This provides **information-theoretic** security for the core audit parameters.

Phase 3: Entanglement-Based Timestamping (Quantum, where available)

- 3.3 Spring and Yajie share entangled pairs for tamper-evident audit record timestamping.
- 3.4 The entanglement witnesses are recorded in the audit ledger as proof of no-tampering.

Phase 4: Classical Digital Signatures for Bulk Data

- 4.4 Bulk audit data is signed with SPHINCS+ (NIST PQC standard, hash-based signatures) or classical Ed25519 (for non-quantum threat models).
- 4.5 All signatures reference the quantum-derived session keys from Phase 2, binding the quantum and classical security domains.

5.5 Cost-Benefit Analysis

Table 3: Cost-Benefit of Quantum Audit / 量子审计成本效益分析

Component / 组件	Unit Cost 单位成本 (USD)	Years to Deploy / 部署年限	Security Gain / 安全增益
BB84 transmitter BB84 发送器	\$5K–\$20K	0–1	High 高
BB84 receiver BB84 接收器	\$10K–\$50K	0–1	High 高
Entanglement source 纠缠源	\$20K–\$100K	1–2	Very High 极高
Quantum memory 量子存储器	\$50K–\$500K	3–5	Very High 极高
Quantum repeater 量子中继器	\$100K–\$1M	5–10	Highest 最高

Cost estimates based on 2024–2026 commercial quantum equipment market.

5.6 Adversary Model and Attack Surface

Definition 5.2 (Quantum Audit Adversary Model). The adversary $\mathcal{A}$ has the following capabilities:

- **Classical:** Unlimited classical computation, full network visibility, ability to inject/modify classical messages.
- **Quantum:** Unlimited quantum computation (fault-tolerant universal quantum computer), ability to perform arbitrary quantum operations on the channel.
- **Physical:** Access to the quantum channel (fiber tapping, free-space interception), but NOT access to the trusted nodes (auditor hardware, ledger hardware).
- **Limitations:** Cannot violate the laws of quantum mechanics (no-cloning, uncertainty principle, causality).

量子审计对手模型：对手拥有无限经典和量子计算能力，可访问量子信道，但不能违反量子力学定律。

Proposition 5.3 (Attack Surface Reduction). *Quantum-secured audit reduces the attack surface from:*

Classical: $\{RSA, ECDSA, AES, SHA-3, TLS, \dots\}$ (surface $\sim$ all of complexity theory)

to:

Quantum: $\{no-cloning, uncertainty principle\}$ (surface $\sim$ 2 physical principles).

This is a reduction from an infinite-dimensional space of potential cryptanalytic attacks to verification of two well-tested physical laws.

6 Integration with the SCX Audit Framework

6.1 Quantum-Secured Audit Protocol (Full Stack)

We present the complete quantum-secured SCX audit protocol, integrating all three quantum primitives into the existing SCX workflow.

Protocol 6.1 (Quantum-Secured SCX Audit (Q-SCX)). **Input:** Data D_t , previous M_t state, auditor configuration.

Output: Audit result $\mathcal{R}_t$ with quantum security guarantees.

Step 1: Quantum Key Establishment

- 1.1 Spring and Yajie each establish BB84-derived keys with the audit ledger: k_S^{BB84} and k_Y^{BB84} .
- 1.2 Spring and Yajie verify shared entanglement via CHSH test (Protocol ??).
- 1.3 They derive an entanglement-based one-time pad k_{OTP} .

Step 2: Quantum-Secured M_t Transmission

- 2.2 Spring computes M_t^{Spring} based on historical consensus.
- 2.3 Yajie computes M_t^{Yajie} based on recent anomaly detection.
- 2.4 Both transmit their M_t vectors to the ledger via BB84 (Protocol ??).
- 2.5 The ledger verifies consistency: if $\|M_t^{\text{Spring}} - M_t^{\text{Yajie}}\| > \delta_{\text{consensus}}$, flag a potential tampering event.

Step 3: Classical SCX Audit with Quantum Binding

- 3.3 The SCX audit proceeds classically (Yajie consensus, Cercis verification, standard expert model queries).
- 3.4 The audit result $\mathcal{R}_t$ is concatenated with a quantum-derived authentication tag:

$$\text{AuthTag}_t = \text{HMAC-SHA3-512}(k_S^{\text{BB84}} \oplus k_Y^{\text{BB84}} \oplus k_{\text{OTP}}, \mathcal{R}_t \parallel t).$$

- 3.5 The tag binds the audit result to the quantum session keys, ensuring that the result cannot be forged without knowledge of the quantum-derived keys.

Step 4: Quantum-Certified Audit Trail

- 4.4 The ledger records $(M_t^{\text{Spring}}, M_t^{\text{Yajie}}, \mathcal{R}_t, \text{AuthTag}_t)$ in the quantum-certified audit log.
- 4.5 Each entry includes the CHSH test statistic S_t as proof of entanglement preservation.
- 4.6 The entanglement witnesses $\{W_t\}$ form an immutable chain of tamper-evident records.

6.2 Security Analysis of Q-SCX

Theorem 6.2 (Q-SCX Security Guarantee). *[RIGOR: FULL] Under the quantum audit adversary model (Definition ??), the Q-SCX protocol (Protocol ??) provides:*

- (i) **M_t Integrity :** *Any adversary that modifies M_t in transit is detected with probability $1 - \text{negl}(\lambda)$.*
- (ii) **Audit Result Authenticity:** *Any adversary that forges $\mathcal{R}_t$ is detected with probability $1 - \text{negl}(\lambda)$.*
- (iii) **Tamper Evidence:** *Any adversary that tampers with the audit trail leaves detectable evidence with probability $1 - \text{negl}(\lambda)$.*

where $\text{negl}(\lambda)$ is a negligible function in the security parameter λ (number of qubits transmitted).

Proof Sketch. (i) M_t is transmitted via BB84 (Protocol ??). By Theorem ??, BB84 provides information-theoretic security: any eavesdropping attempt that extracts non-negligible information introduces an elevated QBER. The protocol aborts when the observed QBER exceeds $Q_{\max} \approx 11\%$; for intercept-resend attacks the expected QBER is 25% (Theorem ??), and the detection probability is $1 - e^{-\Omega(\lambda)}$ by Hoeffding concentration.

(ii) The authentication tag binds $\mathcal{R}_t$ to three independent quantum-derived keys (k_S , k_Y , k_{OTP}). To forge a valid tag, the adversary must know all three keys. Each key is information-theoretically secret (BB84: Theorem ??; OTP: perfect secrecy by entanglement monogamy). The probability of guessing all three keys is $\text{negl}(\lambda)$.

(iii) The audit trail includes CHSH statistics $\{S_t\}$ and entanglement witnesses $\{W_t\}$. By Theorem ??, any tampering that disturbs entanglement by more than $\sim 29\%$ reduces $S \leq 2$ (classical bound). With repeated measurements, the probability of undetected tampering decays as $2^{-\Omega(k)}$, where k is the number of tested Bell pairs. $\square$

6.3 Comparison with Classical Audit Security

Table 4: Classical vs. Quantum Audit Security / 经典与量子审计安全对比

Property / 属性	Classical / 经典	Quantum / 量子
Security basis 安全基础	Computational hardness 计算困难性	Laws of physics 物理定律
Quantum attack resistance 抗量子攻击	No (Shor’s algorithm) 否	Yes, unconditional 是, 无条件安全
Key distribution 密钥分发	Public-key or pre-shared 公钥或预共享	QKD (BB84/Ekert91) 量子密钥分发
Tamper detection 篡改检测	Digital signatures 数字签名	Entanglement witness 纠缠见证
Forward secrecy 前向安全性	Conditional on key erasure 依赖密钥擦除	Guaranteed 有保证
Channel capacity 信道容量	Shannon: $C = B \log_2(1 + \text{SNR})$ 香农定理	HSW: $C_Q = 1 - H_2(2p/3)$ HSW 定理
Hardware cost 硬件成本	$\sim \$0$ (software) $\sim \$0$ (软件)	\$10K – –\$100K \$1万 – –\$10万
Deployment readiness 部署就绪度	Immediate 即时	Partial (co-located) 部分 (共址)

6.4 Transition Roadmap

Phase 1: Immediate (2026–2027): Deploy hybrid classical-quantum audit (Protocol ??) for co-located data centers. Use CRYSTALS-Kyber for key establishment, BB84 for critical M_t parameters, and classical signatures for bulk data.

Phase 2: Near-term (2027–2029): Extend to metro-area audits (< 50 km). Deploy entanglement-based tamper detection between Spring and Yajie. Integrate with quantum random number generators for certified randomness in audit sampling.

Phase 3: Medium-term (2029–2035): Regional deployment contingent on maturation of quantum repeater technology. Full BB84 audit channel with decoy states. Quantum-secured multi-party audit (> 2 auditors). We note that quantum repeaters remain an active research area; deployment timelines depend on breakthroughs in quantum memory coherence times and entanglement purification efficiency.

Phase 4: Long-term (2035+): Global quantum audit network with satellite-based entanglement distribution. Full information-theoretic security for the entire audit pipeline. Integration with quantum internet infrastructure. This phase depends on large-scale quantum network deployment, which realistically extends beyond 2035 based on current progress.

7 Future Directions and Open Problems

7.1 Device-Independent Quantum Audit

Current QKD protocols (including BB84) require trust in the measurement devices. **Device-independent quantum audit** would eliminate this requirement by using Bell

inequalities to certify the quantum nature of the devices. This is the strongest possible security model:

Conjecture 7.1 (Device-Independent Audit Conjecture). *There exists a protocol where the auditor and ledger, using uncharacterized devices, can establish a secure audit channel if and only if the observed Bell violation exceeds a threshold $S > S_{\text{threshold}}$, without trusting any hardware component.*

7.2 Quantum Audit Networks with Multiple Auditors

Generalizing from 2 to $m > 2$ auditors introduces new possibilities:

- **GHZ States:** m -partite entanglement $|\text{GHZ}_m\rangle = (|0\rangle^{\otimes m} + |1\rangle^{\otimes m})/\sqrt{2}$ provides perfect m -party correlation.
- **Quantum Byzantine Agreement:** Achieving consensus among m auditors where up to f may be faulty or compromised, using quantum resources to reduce the classical bound $f < m/3$ to $f < m/2$.
- **Quantum Secret Sharing for Audit Keys:** Splitting the audit key across m auditors such that any k of them can reconstruct it, but fewer than k learn nothing—with information-theoretic security.

7.3 Continuous-Variable Quantum Audit

Discrete-variable QKD (single photons) has low tolerance for loss. **Continuous-variable quantum audit** uses coherent states and homodyne detection, offering compatibility with standard telecom equipment at the cost of more complex error correction:

$$R_{\text{CV-QKD}} \geq \log_2 \left(\frac{V + \chi_{\text{tot}}}{1 + \chi_{\text{tot}}} \right) - \beta I_{AB}$$

where V is the modulation variance and χ_{tot} is the total channel noise.

7.4 Quantum-Secured Federated Audit

For federated learning scenarios where models are audited across multiple data silos:

- **Blind Quantum Audit:** The auditor verifies model properties without learning the model weights (quantum analog of zero-knowledge proofs).
- **Quantum Homomorphic Audit:** Performing audit computations on encrypted quantum data, with the result decrypted only by the authorized recipient.
- **Twin-Field QKD for Star Networks:** Extending audit QKD range via measurement-device-independent techniques in star-topology networks.

7.5 Integration with Blockchain Audit Ledgers

Quantum audit trails can be integrated with blockchain-based audit ledgers for additional immutability guarantees:

- Each block includes a quantum-certified hash of the previous block’s entanglement witness, creating a **quantum blockchain**.
- The no-cloning theorem prevents double-spending attacks on audit records (a quantum record cannot be copied and replayed).

- Quantum timestamping provides provable temporal ordering of audit events with physics-guaranteed accuracy.

8 Conclusion

We have presented a comprehensive formalization of quantum-secured audit for the SCX framework, addressing three fundamental aspects:

- (1) **BB84 for M_t Transmission** : We adapted the BB84 quantum key distribution protocol for secure transmission with theoretical guarantees.
- (2) **Audit Entanglement**: We introduced the concept of audit entanglement, where Spring and Yajie share entangled states that provide tamper detection via CHSH violation tests and entanglement witnesses.
- (3) **Quantum Audit Channel**: We formally defined the quantum audit channel and proved its capacity theorem, showing that the audit information rate is bounded by $C_Q = 1 - H_2(2p/3)$ for the depolarizing channel model.

The key insight is that quantum mechanics elevates audit security from **computational** guarantees (“this is hard to break given current algorithms”) to **physical** guarantees (“this is impossible to break without violating the laws of physics”). This is a qualitative leap in security assurance.

While full quantum-secured audit is not yet deployable at global scale, a hybrid approach combining post-quantum cryptography (CRYSTALS-Kyber, SPHINCS+) with BB84 for critical parameters is feasible today for co-located audit scenarios. We have provided a phased deployment roadmap extending through 2035, aligned with the expected maturation of quantum network infrastructure.

Final Remark / 结语

Quantum mechanics is the strongest security foundation available. It transforms the audit security question from “can the adversary solve this hard problem?” to “can the adversary violate the Heisenberg uncertainty principle?” The answer to the latter is an emphatic no—and to the extent that quantum mechanics is a correct description of physical reality, quantum-secured audit is provably unforgeable.

量子力学是现有最强的安全基础。它将审计安全问题从
“对手能解决这个难题吗？”转变为“对手能违反海森堡不确定性原理吗？”
后者的答案是一个响亮的“不能”——在量子力学是物理现实的正确描述范围内，
量子安全审计是可证明不可伪造的。

A Mathematical Background: Quantum Information Theory

A.1 Density Operators

A quantum state is described by a **density operator** $\rho \in \mathcal{D}(\mathcal{H})$:

$$\mathcal{D}(\mathcal{H}) = \{\rho : \mathcal{H} \rightarrow \mathcal{H} \mid \rho \geq 0, \rho = \rho^\dagger, \text{Tr}(\rho) = 1\}.$$

Pure states: $\rho = |\psi\rangle\langle\psi|$. Mixed states: $\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$.

A.2 Von Neumann Entropy

The von Neumann entropy $S(\rho)$ is the quantum analog of Shannon entropy:

$$S(\rho) = -\text{Tr}(\rho \log_2 \rho) = -\sum_i \lambda_i \log_2 \lambda_i,$$

where $\{\lambda_i\}$ are the eigenvalues of ρ .

A.3 Fidelity

The fidelity between two states ρ and σ :

$$(\rho, \sigma) = \left(\text{Tr} \sqrt{\sqrt{\rho} \sigma \sqrt{\rho}} \right)^2.$$

For pure states: $(|\psi\rangle, |\phi\rangle) = |\langle\psi|\phi\rangle|^2$.

A.4 CHSH Inequality (Bell's Theorem)

For measurements A_0, A_1 (Alice) and B_0, B_1 (Bob) with outcomes ± 1 , the CHSH value:

$$S = \mathbb{E}[A_0 B_0] + \mathbb{E}[A_0 B_1] + \mathbb{E}[A_1 B_0] - \mathbb{E}[A_1 B_1].$$

Local-hidden-variable theories satisfy $|S| \leq 2$ (CHSH inequality). Quantum mechanics allows $|S| \leq 2\sqrt{2}$ (Tsirelson bound). Entanglement is certified when $2 < |S| \leq 2\sqrt{2}$.

A.5 Holevo Bound

The Holevo quantity bounds the accessible classical information in a quantum ensemble $\{p_x, \rho_x\}$:

$$\chi = S\left(\sum_x p_x \rho_x\right) - \sum_x p_x S(\rho_x).$$

This is the classical capacity of a noiseless quantum channel.

B Proof of the Audit Channel Capacity Lower Bound

Full Proof of Theorem ??. For the depolarizing channel $\mathcal{N}_p$ with parameter $p \in [0, 1]$:

$$\mathcal{N}_p(\rho) = (1-p)\rho + \frac{p}{3}(\sigma_x \rho \sigma_x + \sigma_y \rho \sigma_y + \sigma_z \rho \sigma_z).$$

Step 1: Channel symmetry. The depolarizing channel is covariant under the Pauli group:

$$\mathcal{N}_p(U\rho U^\dagger) = U\mathcal{N}_p(\rho)U^\dagger$$

for any single-qubit unitary U . This implies that the optimal input ensemble is the uniform distribution over a complete set of mutually unbiased bases (e.g., the six states $\{|0\rangle, |1\rangle, |+\rangle, |-\rangle, |+i\rangle, |-i\rangle\}$).

Step 2: Output entropy. For the maximally mixed input $\rho_{\text{avg}} = \mathbb{I}/2$:

$$\mathcal{N}_p(\mathbb{I}/2) = (1-p)\frac{\mathbb{I}}{2} + \frac{p}{3} \cdot 3 \cdot \frac{\mathbb{I}}{2} = \frac{\mathbb{I}}{2}.$$

Thus $S(\mathcal{N}_p(\mathbb{I}/2)) = S(\mathbb{I}/2) = \log_2 2 = 1$.

Step 3: Conditional output entropy. For a pure state input $\rho = |\psi\rangle\langle\psi|$, choose $|\psi\rangle = |0\rangle$ without loss of generality (by covariance):

$$\mathcal{N}_p(|0\rangle\langle 0|) = (1-p)|0\rangle\langle 0| + \frac{p}{3}(|1\rangle\langle 1| + |1\rangle\langle 1| + |0\rangle\langle 0|) = \left(1 - \frac{2p}{3}\right)|0\rangle\langle 0| + \frac{2p}{3}|1\rangle\langle 1|.$$

This is a classical mixture with eigenvalues $\lambda_0 = 1 - 2p/3$ and $\lambda_1 = 2p/3$. The von Neumann entropy of this output state is therefore:

$$S(\mathcal{N}_p(|\psi\rangle\langle\psi|)) = H_2\left(\frac{2p}{3}\right) = -\frac{2p}{3}\log_2\frac{2p}{3} - \left(1 - \frac{2p}{3}\right)\log_2\left(1 - \frac{2p}{3}\right).$$

Step 4: Holevo quantity. The Holevo quantity for our p -parametrization is:

$$\chi(\mathcal{N}_p) = S(\mathcal{N}_p(\mathbb{I}/2)) - S(\mathcal{N}_p(|\psi\rangle\langle\psi|)) = 1 - H_2\left(\frac{2p}{3}\right).$$

Step 5: Additivity. For the depolarizing channel, the Holevo quantity is additive: $\chi(\mathcal{N}_p^{\otimes n}) = n\chi(\mathcal{N}_p)$ [?]. Hence:

$$C(\mathcal{N}_p) = \chi(\mathcal{N}_p) = 1 - H_2\left(\frac{2p}{3}\right).$$

Numerical check (using our p -parametrization):

- $p = 0$: $C_Q = 1 - H_2(0) = 1.000$ (noiseless, 1 bit/qubit).
- $p = 0.11$: $C_Q = 1 - H_2(0.0733) \approx 0.622$ bits/qubit.
- $p = 0.25$: $C_Q = 1 - H_2(0.1667) \approx 0.350$ bits/qubit.
- $p = 0.75$: $C_Q = 1 - H_2(0.5) = 0.000$ (capacity zero).

□

C Implementation Notes for Q-SCX Prototype

C.1 Software Stack

Quantum-Secured SCX Audit — Software Stack

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Layer 1: Quantum Hardware Abstraction

- Qiskit / Cirq / PennyLane for quantum circuit simulation
- QKD vendor SDKs (ID Quantique, QuintessenceLabs, Toshiba)
- Quantum random number generator API

Layer 2: Quantum Protocol Engine

- BB84 state machine (prepare, transmit, measure, sift)
- Cascade error correction (BB84 reconciliation)
- Toeplitz hashing (privacy amplification)
- CHSH test executor (entanglement verification)

Layer 3: SCX Audit Integration

- Mt quantization and encoding
- Quantum-derived key management

- Authentication tag generation and verification
- Audit trail entanglement witness recorder

Layer 4: Classical Fallback

- CRYSTALS-Kyber (liboqs / OpenQuantumSafe)
- SPHINCS+ signatures
- Classical audit pipeline (unchanged SCX core)

C.2 Key Parameters

Table 5: Recommended Security Parameters / 推荐安全参数

Parameter / 参数	Value / 值	Rationale / 理由
BB84 photon count n	$\geq 10^4$	$> 2 \times$ minimal for 256-bit key 256 位密钥最小值的 2 倍以上
Error estimation sample k	$n/4$	Statistical significance 统计显著性
QBER threshold $Q_{\max}$	0.08–0.11	Conservative
Privacy amplification output	256 bits	Matches AES-256 security 匹配 AES-256 安全级别
Entanglement test pairs t	1000	Detects $\delta > 0.01$ with > 0.9999 prob. 以 > 0.9999 概率检测 $\delta > 0.01$
CHSH threshold	2.2	Margin above classical bound of 2
Kyber security level	ML-KEM-1024	NIST Level 5 (highest) NIST 第 5 级 (最高)

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